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Attack detection and fusion-based security state estimation in multi-sensor networks

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ABSTRACT

This study aims to address the challenges of attack detection and sensor data fusion in multi-sensor stochastic systems where multi-sensor networks are subjected to cyber threats. From the standpoint of minimizing output error power, a detection method is proposed to monitor sensor transmission channels for possible attacks, where the warning system is modeled as a binary hypothesis test. Based on the detection algorithm, the attacked sensor around the fusion center is found, and the attacked sensor signal is filtered by the idea of s -sparse observability. Then, the optimal fusion weight matrix is recalculated by minimizing the error between the fusion center and adjacent sensors and using the Lagrange multiplier method to reconstruct the sensor fusion strategy. Research results confirm that the technique reliably maintains fusion accuracy and robustness in attack scenarios. Through an application in radar target tracking, the secure state estimation results illustrate the method's practical utility.

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1. Introduction

Cyber-physical systems (CPSs) constitute sophisticated frameworks merging computing capabilities, communication networks, and physical operations. Advances in ambient sensing technologies – including wireless sensor networks and computer vision – have enabled widespread deployment of CPSs in autonomous vehicles (Xiong et al., 2015), smart grid (Lin, 2013), and industrial automation (Du et al., 2019). Recent years have witnessed a dramatic rise in cyber attacks, raising serious concerns about CPSs security and operational reliability. In the context of smart grids, malevolent operations have the potential to significantly disrupt the power supply, thereby potentially leading to widespread power outages. The water treatment poisoning (Ali et al., 2018) and Ukraine's grid cyberattack (Luo et al., 2024) demonstrate how network infiltration and/or data tampering compromise CPSs security. Therefore, the research on the security of CPSs is crucial.

Malicious cyber operations against CPSs manifest as either stealthy or non-stealthy attack modalities. Stealthy attacks are specifically designed to infiltrate

the system covertly, aiming to manipulate CPSs without triggering any alarms or detection. The attack strategies in this category primarily include False Data Injection (FDI) attacks (Shi et al., 2017; Sun et al., 2023; Yi & Xu, 2024), which enable attackers to cause long-term damage unnoticed. In contrast, unstealthy attacks are overt in nature and typically result in immediate system anomalies, such as Denial of Service (DoS) attacks (X. Liu et al., 2022; W. Liu et al., 2023; Lu & Yang, 2020; L. Wang et al., 2024; Zhao & Zhu, 2021) and partial feedback attacks (Shi et al., 2017).

Although defending against such stealthy attacks is challenging, defenders have strategies to deal with them. From the perspective of defender, the integration of attack detection and security state estimation algorithms is an effective strategy, particularly in the context of FDI attacks. The detection of attacks poses a fundamental challenge in securing CPSs. The detection of FDI attacks has garnered significant attention in recent years (Lu & Yang, 2023; Suo et al., 2024b; Zhao et al., 2023). The χ^2 detector, utilizing residual signals for attack identification, constitutes the most frequently investigated technique in current research (G.

Chen et al., 2022). Incorporating historical residuals, Ye and Zhang (2020) improved the χ^2 detector's accuracy and broadened its detection range. Leveraging existing research, Y. Wang et al. (2023) devised an attack detection scheduling strategy aimed at enhancing the efficiency of distributed detection. The process of selecting suspicious sensors, however, involves a significant element of randomness. In recent years, Suo et al. (2024a) researched an active detection scheme based on trusted nodes. The design of active detection methods in distributed networks, however, is a complex task due to the inherent difficulty in acquiring advance attack information. Therefore, developing attack detection algorithms for inter-sensor communication in distributed networks presents a significant research challenge.

If a network attack is detected in the system, how to reduce or eliminate the impact on the system is a hot topic in the field of secure CPSs. By providing security state estimation, the detrimental effects of malicious attacks on systems can be mitigated. The proliferation of sensors in distributed networks has driven extensive research into state estimation leveraging information redundancy. Previous work by Shoukry et al. (2017) investigated attacks in sensor networks, establishing an upper bound on the number of malicious (or compromised) sensors the system can tolerate. To improve system robustness against malicious sensors, Su et al. (2023) leveraged sparse initial state information. Sensor fusion has recently expanded into state estimation, driven by the proliferation of wireless sensor networks. Shen et al. (2023) developed an error-based weighted fusion strategy to mitigate the adverse effects of lossy sensors on system performance. In multi-sensor networks, a single sensor cannot fully observe the state of the system; therefore, multiple sensors need to work together to perform complete state estimation. Current research often addresses linear compensation for lossy channels (M. Chen et al., 2011; Chen & Li, 2021; Suo et al., 2024a) but rarely considers FDI attacks. Therefore, the development of a sensor fusion algorithm that can maintain the fusion accuracy and effectively defend against FDI attacks is helpful to the safe and stable operation of the system. Consequently, this paper addresses the challenge of designing effective sensor fusion algorithms for stochastic systems subject to FDI attacks.

This paper proposes a strategy integrating attack detection with sensor fusion for stochastic systems under network attacks. Bounded state estimation is achievable provided the number of sensors connected to the fusion center remains below the sparsity limit.

- (1) Aiming at the communication security problem between sensor networks, an attack detection algorithm based on average output error is designed. Then, according to the principle of attack detection and prediction, the early warning mechanism is reflected by the method of hypothesis testing.
- (2) Under the premise of the detector, the attacked sensor is processed by sparse principle in fusion and then weighted fusion estimation is carried out. This fusion strategy greatly improves the computational efficiency without affecting the estimation accuracy.
- (3) Theoretical analysis and simulation results demonstrate that the proposed joint detection-fusion framework achieves secure state estimation for stochastic systems in multi-sensor networks, provided the number of compromised sensors satisfies the sparsity constraint.

The remaining sections of this paper are as follows. Section 2 provides an overview of the problem. The design of an attack signal detector is presented in Section 3. In Section 4, the design and effect derivation of a security sensor fusion algorithm against network attacks are introduced in detail. Section 5 offers simulation examples. Finally, Section 6 concludes the paper.

Notations: The symbols used in the formulas of this paper are as follows. $\mathbb{R}$ denotes the set of real numbers, and $\mathbb{R}^n$ represents n -dimensional Euclidean space. A^T and $\text{tr}(A)$ denote the transpose and trace of matrix A , respectively. For the matrix $A > 0$, it indicates that A is positive definite. $|N|$ denotes the cardinality of set N . N/M is the remaining subset of the set N after removing the subset M . For vector a , $|a| = \sqrt{a^T a}$. $E(\cdot)$ denotes expectation. $\mathcal{N}(\mu, \sigma^2)$ denotes a Gaussian distribution with mean μ and variance σ^2 . I_n denotes the n -dimensional identity matrix.

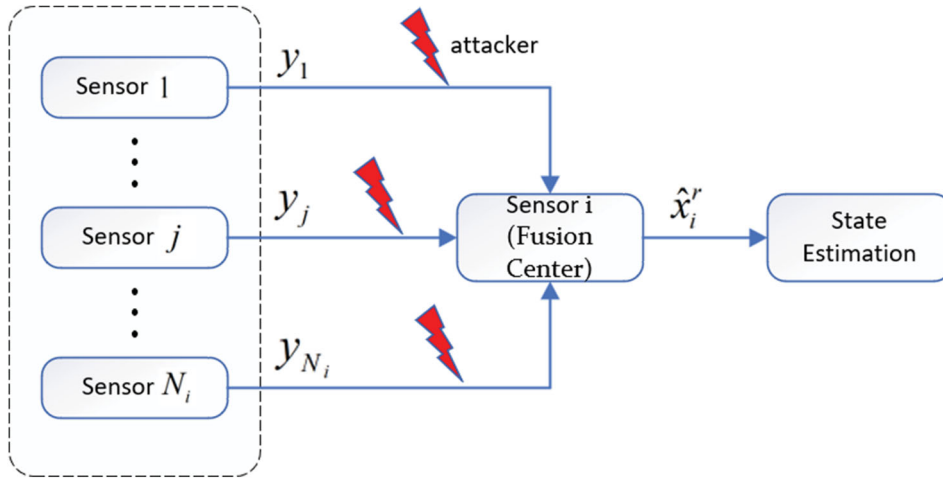


Figure 1. Insecure local multi-sensor network diagram.

2. Problem statement

2.1. System model

This paper considers a distributed sensor network described as $G = (N, \theta)$. N and $\theta \subseteq N \times N$ represent the set of fusion centers and the total set of adjacent sensors, respectively. $N_i = \{j \in N: (i, j) \in \theta\}$ represents the set of sensors adjacent to the i th sensor. The local sensor network appears in Figure 1. The achievement of accurate state estimation necessitates that each sensor within the sensor network operates as a fusion center for fusion estimation. Consider a discrete linear time-varying system

$$x(k+1) = A(k)x(k) + B(k)u(k) + \omega(k) \quad (1)$$

where $x(k) \in \mathbb{R}^n$, $u(k) \in \mathbb{R}^p$, and $\omega(k) \in \mathbb{R}^n$ represent the states of the system, control inputs, and process noise, respectively. $\omega(k)$ obeys the Gaussian distribution of zero mean and covariance matrix $Q(k) > 0$, i.e. $\omega(k) \sim \mathcal{N}(0, Q(k))$.

For the i th sensor, the measurement model is

$$y_i(k) = C_i(k)x(k) + v_i(k) \quad (2)$$

where $y_i(k) \in \mathbb{R}^m$ and $v_i(k) \in \mathbb{R}^m$ represent the output of the i th sensor and the measurement noise, respectively. In addition, $v_i(k)$ follows $\mathcal{N}(0, R_i(k))$, where $R_i(k) > 0$.

Assumption 2.1: The control system parameter matrices $A(k)$, $B(k)$ and $Q(k)$ satisfy that $(A(k), B(k))$ is controllable, $(A(k), C_i(k))$ is observable, and $(A(k), \sqrt{Q(k)})$ is stabilizable.

Assumption 2.2 (Sun & Deng, 2004): The initial state $x(0)$, $\omega(k)$ and $v_i(k)$, $i = 1, 2, \dots, m$ are independent of each other.

Remark 2.1: The first two conditions in Assumption 2.1 are the basic conditions for the stability of the Kalman filter, and the latter is a prerequisite for s -sparse observability (Mishra et al., 2017). The initial state of the system can be set, and different noises are not related, because different sensors are distributed at different locations and the interference noise is naturally different (Sun & Deng, 2004). Therefore, the establishment of hypothesis 2.2 is reasonable.

2.2. State estimation

Similar to Suo et al. (2024a), each sensor is treated in the sensor network as a fusion center, which can receive and integrate information by neighbouring sensors. At moment $k+1$, the fusion center fuses the state estimation $\hat{x}_i(k+1)$ of itself and the state estimation $\hat{x}_{ij}(k+1)$, $j \in N_i$ of adjacent sensors by weighted fusion.

The estimator for the fusion center i is given as

$$\hat{x}_i(k+1) = \hat{x}_i^{r-}(k+1) + Z_i(k+1) \quad (3)$$

where $\hat{x}_i^{r-}(k+1) = \hat{x}_i^r(k) + B(k)u(k)$, $Z_i(k+1) = K_i(k+1)(y_i(k+1) - C_i(k+1)\hat{x}_i^{r-}(k+1))$, where $K_i(k+1)$ is the gain matrix of the i th sensor. $Z_i(k+1)$ represents the innovation of Kalman filter.

The fused sensor estimate $\hat{x}_i^r(k+1)$ is derived from

$$\hat{x}_i^r(k+1) = \sum_{j \in N_i} W_{ij}(k+1)\hat{x}_j(k+1) \quad (4)$$

where $W_{ij}(k+1)$ represents the weighted coefficient matrix between sensor i and its neighbour j . The fusion estimation at the next time step will be influenced by the determination of fusion coefficient matrix, as indicated in Equation (4).

Exploitation risks stemming from the unsecured medium of wireless sensor networks necessitate examination of attack vectors on inter-node links. This paper consequently focuses on modeling compromised signals within sensor communication pathways, as visualized in Figure 1. Specifically, adversarial data $z_{ij}(k)$ infiltrates sensor transmit signals $y_j(k)$ while traversing the (i, j) network channel, that is

$$y_{ij}^a(k) = y_j(k) + z_{ij}(k) \quad (5)$$

where $y_{ij}^a(k)$ represents the tampered data transmitted from the j th sensor to the fusion center.

For the subsequent detector design, the concept of average malicious disturbance power (AMDP) is introduced, the following lemma is shown.

Lemma 2.1 (Fang, 2021): *As time k approaches infinity, the AMDP of the attack signal $z_{ij}(k)$ injected into the measurement signal of the j th sensor is defined as $P_{z_{ij}}$, where $t = 1:k$, that is,*

$$P_{z_{ij}} = \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|z_{ij}(t)\|^2$$

Remark 2.2: On the one hand, under the condition that the j th sensor remains unattacked, $z_{ij}(k) = 0$. Therefore, $P_{z_{ij}} = 0$ is real before any attack by the adversary. On the other hand, when the attack signal exists, $z_{ij}(k)$ cannot always be zero, then $P_{z_{ij}} > 0$ is always satisfied.

3. Detection strategy design

To mitigate against the unpredictable effects of malicious attacks on the sensor fusion process, every communication channel depicted in Figure 1 requires the integration of an attack detection mechanism. Leveraging the dynamic evolution characteristics of AMDP, this section designs and implements a real-time attack detection algorithm.

Definition 3.1: At time k , The average output error power (AOEP) is denoted as Γ_{ij}^w for the unattacked

communication channel between the j th and i th sensors, and Γ_{ij}^a for its attacked counterpart, i.e.

$$\begin{cases} \Gamma_{ij}^w = \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|y_j(t) - y_i(t)\|^2, & z_{ij}(k) = 0 \\ \Gamma_{ij}^a = \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|y_j^a(t) - y_i(t)\|^2, & z_{ij}(k) \neq 0 \end{cases} \quad (6)$$

Remark 3.1: Suppose that the system is normal during the start of operation, then the AOEP at $z_{ij}(k) = 0$ in (6) can be calculated at each time according to the output errors of the j th and i th sensors in the past period of time. This establishes a baseline Γ_{ij}^w for normal operation, enabling the detector to identify attacks through significant deviations in the observed AOEP, as formalised by the hypothesis test in (7).

Theorem 3.1: *During the operation when the system has not been attacked, the value of Γ_{ij}^w is calculated. Based on Equation (6), when the j th sensor is attacked, Γ_{ij}^w is immediately transformed into Γ_{ij}^a and the relationship between them can be obtained, i.e.*

$$\Gamma_{ij}^a > \Gamma_{ij}^w$$

Proof: In order to facilitate the subsequent calculation, define

$$\begin{aligned} X_\omega &= \lim_{k \rightarrow \infty} \text{cov} \left(\sum_{t=0}^{k-1} \prod_t^{k-t-1} A(t+1) \omega(t), \right. \\ &\quad \left. \sum_{t=0}^{k-1} \prod_t^{k-t-1} A(t+1) \omega(t) \right) \\ X_a &= \lim_{k \rightarrow \infty} \text{cov} \left(\sum_{t=0}^{k-1} \prod_t^{k-t-1} A(t+1) B(t) u(t), \right. \\ &\quad \left. \sum_{t=0}^{k-1} \prod_t^{k-t-1} A(t+1) B(t) u(t) \right) \end{aligned}$$

The recursions of X_a and X_ω are shown as follows:

$$\begin{aligned} X_\omega &= \sum_{t=0}^{\infty} \prod_t^{k-t-1} A(t+1) \text{cov}(\omega(t), \omega(t)) \\ &\quad \times \left(\prod_t^{k-t-1} A(t+1) \right)^T \end{aligned}$$

$$\begin{aligned}
&= A(t+1)X_\omega A^T(t+1) + Q(t) \\
X_a &= \sum_{t=0}^{\infty} \prod_t^{k-t-1} A(t+1) \text{cov}(B(t)u(t), B(t)u(t)) \\
&\quad \times \left(\prod_t^{k-t-1} A(t+1) \right)^T \\
&= A(t+1)X_u A^T(t+1) \\
&\quad + \text{cov}(B(t)u(t), B(t)u(t))
\end{aligned}$$

Then, $z_{ij}(k) = 0$, the derivation of Γ_{ij}^w is as follows:

$$\begin{aligned}
\Gamma_{ij}^w &= \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|y_j(t) - y_i(t)\|^2 \\
&= \lim_{k \rightarrow \infty} \text{cov}((y_j(k) - y_i(k)), (y_j(k) - y_i(k))) \\
&= \lim_{k \rightarrow \infty} \text{cov}((C_j(k)x_j(k) + v_j(k) - (C_i(k)x_i(k) \\
&\quad + v_i(k)), C_j(k)x_j(k) + v_j(k) \\
&\quad - (C_i(k)x_i(k) + v_i(k))) \\
&= \lim_{k \rightarrow \infty} \text{cov}((C_j(k)x_j(k), C_j(k)x_j(k)) + \text{cov}(C_j(k) \\
&\quad \times x_j(k), v_j(k)) - \text{cov}(C_j(k)x_j(k), C_i(k)x_i(k)) \\
&\quad - \text{cov}(C_j(k)x_j(k), v_i(k)) \\
&\quad + \text{cov}(v_j(k), C_j(k)x_j(k)) \\
&\quad + \text{cov}(v_j(k), v_j(k)) - \text{cov}(v_j(k), C_i(k)x_i(k)) \\
&\quad - \text{cov}(v_j(k), v_i(k)) \\
&\quad - \text{cov}(C_i(k)x_i(k), C_j(k)x_j(k)) \\
&\quad - \text{cov}(C_i(k)x_i(k), v_j(k)) + \text{cov}(C_i(k)x_i(k), \\
&\quad C_i(k)x_i(k)) + \text{cov}(C_i(k)x_i(k), v_i(k)) \\
&\quad - \text{cov}(v_i(k), C_j(k)x_j(k)) - \text{cov}(v_i(k), v_j(k)) \\
&\quad + \text{cov}(v_i(k), C_i(k)x_i(k)) + \text{cov}(v_i(k), v_i(k)) \\
&= \lim_{k \rightarrow \infty} \text{cov}((C_j(k)x_j(k), C_j(k)x_j(k)) \\
&\quad + \text{cov}(C_i(k)x_i(k), C_i(k)x_i(k)) + R_i + R_j
\end{aligned}$$

Since $x(0)$ and $u(k)$ are known, they can be set to zero, and assuming the spectral radius of $A(k)$ is no more than 1, then the following results can be obtained.

$$\begin{aligned}
\Gamma_{ij}^w &= \lim_{k \rightarrow \infty} \text{cov}((C_j(k)x_j(k), C_j(k)x_j(k)) \\
&\quad + \text{cov}(C_i(k)x_i(k), C_i(k)x_i(k)) + R_i + R_j
\end{aligned}$$

$$\begin{aligned}
&= \lim_{k \rightarrow \infty} \text{cov}(C_j(k) \prod_t^k A(t)x_j(0), C_j(k) \\
&\quad \times \prod_t^k A(t)x_j(0)) + \text{cov}(C_j(k) \\
&\quad \sum_{t=0}^{\infty} \prod_t^{k-t-1} A(t+1)\omega(t), C_j(k) \sum_{t=0}^{\infty} \prod_t^{k-t-1} \\
&\quad A(t+1)\omega(t)) + \text{cov}(C_j(k) \sum_{t=0}^{\infty} \prod_t^{k-t-1} A(t+1) \\
&\quad \times B(t)u(t), C_j(k) \sum_{t=0}^{\infty} \prod_t^{k-t-1} A(t+1)B(t)u(t)) \\
&\quad + \text{cov}(C_i(k) \prod_t^k A(t)x_i(0), C_i(k) \prod_t^k A(t)x_i(0)) \\
&\quad + \text{cov}(C_i(k) \sum_{t=0}^{\infty} \prod_t^{k-t-1} A(t+1)\omega(t), C_i(k) \\
&\quad \times \sum_{t=0}^{\infty} \prod_t^{k-t-1} A(t+1)\omega(t)) \\
&\quad + \text{cov}(C_i(k) \sum_{t=0}^{\infty} \prod_t^{k-t-1} A(t+1)B(t)u(t), \\
&\quad \times C_i(k) \sum_{t=0}^{\infty} \prod_t^{k-t-1} A(t+1)B(t)u(t)) + R_i + R_j \\
&= C_j X_\omega C_j^T + C_j X_a C_j^T + C_i X_\omega C_i^T \\
&\quad + C_i X_a C_i^T + R_i + R_j
\end{aligned}$$

Similarly, when under attack, i.e. $z_{ij}(k) \neq 0$

$$\begin{aligned}
\Gamma_{ij}^a &= \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|y_j^a(t) - y_i(t)\|^2 \\
&= \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|y_j^a(t) + z_{ij}(t) - y_i(t)\|^2 \\
&= \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|y_j^a(t) + z_{ij}(t)\|^2 - 2((y_j^a(t) \\
&\quad + z_{ij}(t))y_i(t) + \|y_i(t)\|^2 \\
&\stackrel{(a)}{=} \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|y_j^a(t) + z_{ij}(t)\|^2 + \|y_i(t)\|^2
\end{aligned}$$

$$\begin{aligned}
&= \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|y_j^a(t)\|^2 + \|z_{ij}(t)\|^2 + \|y_i(t)\|^2 \\
&= \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|y_j(t) - y_i(t)\|^2 \\
&\quad + \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{t=1}^k \|z_{ij}(t)\|^2 \\
&= C_j X_\omega C_j^T + C_j X_a C_j^T + C_i X_\omega C_i^T + C_i X_a C_i^T \\
&\quad + R_i + R_j + P_{z_{ij}} > \Gamma_{ij}^w
\end{aligned}$$

This proof is completed. $\blacksquare$

Remark 3.2: The equation (a) remains valid due to the fact that the covariance between uncorrelated variables is zero. In this case, the method of M. Chen et al. (2011) can be used to design adaptive control under certain conditions to achieve the constraint that the spectral radius of the system does not exceed 1. In practise, this condition corresponds to stable or critical stable system dynamics, which is the intrinsic physical property of most actual controlled objects (such as mechanical and chemical processes). Therefore, it is reasonable to meet the conditions for the stable operation of most real systems.

Based on Theorem 3.1 and Remark 3.1, the conclusion can be drawn that if the transmission measurement signal from the j th sensor to the i th sensor remains unaffected by the attack, the average measurement error will remain unchanged at Γ_{ij}^w . The occurrence of an attack at any time will result in the transition of Γ_{ij}^w to Γ_{ij}^a . Therefore, whether an attack occurs between the j th sensor to the i th sensor at time k can be framed as a hypothesis testing problem, i.e.

$$\begin{cases} H_0^{ij}(k) : \frac{\Gamma_{ij}^w}{\Gamma_{ij}^w} = 1, & \text{without attack} \\ H_1^{ij}(k) : \frac{\Gamma_{ij}^a}{\Gamma_{ij}^w} > 1, & \text{with attack} \end{cases} \quad (7)$$

If $H_1^{ij}(k)$ in (7) is satisfied at time k , the j th sensor is determined to be attacked and an alarm is raised. When Γ_{ij}^a remains time-invariant for significantly extended periods, the threshold Γ_{ij}^w can be updated to the current value Γ_{ij}^a , which means that (7) can detect finite-time attacks.

Remark 3.3: The effective operation of the proposed detection algorithm relies on the premise that the system must run normally for a certain duration. However, this scenario is highly prevalent in real-world situations due to the attacker's lack of knowledge regarding the system's operational periods. Since the energy of the attack signal is limited, it is reasonable for (7) to detect finite-time attacks.

4. Secure sensor fusion

The main focus of this paper is to achieve a flawless defence against network attacks by combining the proposed attack detection algorithm with security state estimation. Consequently, this section focuses on devising a fusion algorithm that synthesises state estimations from both the fusion center and neighbouring sensors to attain high-fidelity system state reconstruction. Theoretically, the algorithm can achieve secure state estimation for each fusion center if the number of sensors surrounding it satisfies the condition of s -sparse observability. Some concepts and explanations related to s -sparse observability are given below.

4.1. S -sparse observability

Definition 4.1 (Mishra et al., 2017): For the sensor set D and $M \subseteq D$, if $|M| \leq s$ sensors are damaged, but $(A, C_{D/M})$ is still observable, then it can be called s -sparse observability.

Remark 4.1: The paper establishes set D to encompass all sensors surrounding the fusion center, while set M represents the subset of attacked sensors. Consequently, the network surrounding the fusion center exhibits s -sparse observability. This implies that as long as the number of attacked sensors in the adjacent set does not exceed s , a secure state estimation can be achieved through utilization of the sensor fusion algorithm.

4.2. The design of local estimator

By combining the system model and the fusion center estimator, the state estimator of the j th sensor can be obtained. The recursion of the specific local estimator is shown as follows:

$$\hat{x}_j(k+1) = \hat{x}_j^-(k+1) + Z_j(k+1) \quad (8)$$

$$K_j(k+1) = P_j^-(k+1)C_j^T(k+1)[C_j(k+1)P_j^-(k+1)C_j^T(k+1) + R_j(k+1)]^{-1} \quad (9)$$

$$P_j^-(k+1) = A(k)P_j(k)A^T(k) + Q(k) \quad (10)$$

$$P_j(k+1) = (I + K_j(k+1)C_j(k+1))P_j^-(k+1) \quad (11)$$

where $P_j^-(k+1)$ is derived from the previous time step's self-covariance matrix and the system coefficient matrix, and $P_j(k+1)$ is the autocovariance matrix at time $k+1$.

Remark 4.2: The recursive form of the local estimator for the j th sensor corresponds to the fundamental recursive structure of the Kalman filter, thus its proof is omitted in this context. Notably, each sensor can be used as a fusion center, so the structure of the local estimator is similar to that of (3).

4.3. Fusion algorithm under attack signal

To improve the transmission efficiency and realise the global state estimation, it is necessary to fuse the local state estimation with the fusion center state estimation. However, the potential risk of attacks that may lead to data tampering is present when measurement information is transmitted to the fusion center by adjacent sensors. Therefore, a fusion algorithm is proposed in this section to mitigate the impact of attack signals. Provided the number of compromised sensors surrounding the fusion center remains within the prescribed threshold, the algorithm yields secure state estimates for each fusion node.

Theorem 4.1: Under the condition of Assumption 2.2, when the transmission channel between the j th sensor and the i th sensor is not destroyed, the error cross-covariance $P_{ij}(k+1)$ of the local Kalman filter has the following recursive formula:

$$\begin{aligned} P_{ij}(k+1) &= [I_n - K_j(k+1)C_j(k+1)][A(k)P_{ij}(k)A^T(k) \\ &\quad + Q(k)][I_n - K_j(k+1)C_j(k+1)]^T \\ &\quad + K_j(k+1)S_{ij}(k+1)K_j(k+1)^T \end{aligned} \quad (12)$$

where $P_{ij}(k)$ and $S_{ij}(k+1) = E(v_i(k+1)v_j^T(k+1))$ are the cross-covariance matrices of local error cross-covariance and measurement disturbance at the previous moment, respectively.

Proof: $\Delta\hat{x}_j(k+1) = x(k+1) - \hat{x}_j(k+1)$ is defined as the estimation error of $k+1$ at time in the transmission channel.

$$\begin{aligned} \Delta\hat{x}_j(k+1) &= A(k)x(k) + B(k)u(k) + \omega(k) - \hat{x}_j^-(k+1) \\ &\quad - K_j(k+1)[y_j(k+1) - \hat{x}_j^-(k+1)] \\ &= [I_n - K_j(k+1)C_j(k+1)][A(k)\Delta\hat{x}_j(k) \\ &\quad + \omega(k)] - K_j(k+1)v_j(k+1) \end{aligned} \quad (13)$$

Based on (13), the following recursive formula can be obtained:

$$\begin{aligned} P_{ij}(k+1) &= E(\Delta\hat{x}_i(k+1)\Delta\hat{x}_j^T(k+1)) \\ &= E(\{[I_n - K_j(k+1)C_j(k+1)][A(k)\Delta\hat{x}_{ij}(k) \\ &\quad + \omega(k)] - K_j(k+1)v_j(k+1)\}\{[I_n - K_j(k+1) \\ &\quad C_j(k+1)][A(k)\Delta\hat{x}_{ij}(k) + \omega(k)] \\ &\quad - K_j(k+1)v_j(k+1)\}^T) \\ &= [I_n - K_j(k+1)C_j(k+1)][A(k)P_{ij}(k) \\ &\quad A^T(k) + Q(k)][I_n - K_j(k+1)C_j(k+1)]^T \\ &\quad + K_j(k+1)S_{ij}(k+1)K_j(k+1)^T \end{aligned} \quad (14)$$

This proof is completed. ■

Remark 4.3: According to Assumption 2.2, $\omega(k)$, $v_i(k+1)$ and $v_j(k+1)$ and the past estimation errors $\Delta\hat{x}_i(k)$ and $\Delta\hat{x}_j(k)$ are independent of each other. This means that the expected value of any product containing one noise term and another irrelevant noise term or state error term will be zero.

For example:

$$\begin{aligned} E[\omega(k)v_j^T(k+1)] &= E[\Delta\hat{x}_i(k)v_j^T(k+1)] \\ &= E[v_i(k+1)\omega^T(k)] \\ &= E[v_i(k+1)\Delta\hat{x}_j^T(k)] = 0 \end{aligned}$$

Therefore, after further simplification, the results of Equation (14) can be obtained.

Considering the potential adversarial compromise of sensor networks, lemma 4.1 lays a theoretical foundation for constructing the optimal sensor fusion framework.

Lemma 4.1 (Sun & Deng, 2004): Denote $\hat{x}_j(k+1)$ as the unbiased state estimation of n -dimensional vector $x_j(k+1)$, and the auto-covariance matrix and the cross-covariance matrix are represented by $P_i(k+1)$ and $P_{ij}(k+1)$, respectively. The form of fusion estimator with weight matrix is as follows:

$$\begin{aligned} \hat{x}_i^r(k+1) = & w_{i1}\hat{x}_{i1}(k+1) + \cdots + w_{ij}\hat{x}_{ij}(k+1) \\ & + \cdots + w_{i|N_i|}\hat{x}_{i|N_i|}(k+1) \end{aligned} \quad (15)$$

where w_{ij} is the coefficient matrix of the corresponding state estimation.

Remark 4.4: All the coefficient matrices in (15) satisfy $w_{i1} + w_{i2} + \cdots + w_{iN_i} = I_n$. Considering that the j th sensor may be attacked when transmitting information, it is necessary to further design the optimal fusion algorithm.

The cross-covariance matrix Ξ of the fusion estimator can be calculated by combining (12) and Lemma 4.1, that is,

$$\begin{aligned} \Xi &= E(\Delta\hat{x}_{ij}(k+1)\Delta\hat{x}_{ij}^T(k+1)) \\ &= W_{ij}(k+1)P_{ij}(k+1)W_{ij}^T(k+1) \end{aligned} \quad (16)$$

where $W_{ij}(k+1) = (w_{i1}, w_{i2}, \dots, w_{i|N_i|})_{n|N_i| \times n}^T$.

Before calculating the optimal fusion matrix, the attack detection algorithm proposed in this paper is used to determine whether the j th sensor is attacked.

Case 1: The j th sensor is normal. The method of Lemma 4.1 is directly used for information fusion.

Case 2: The j th sensor is under attack. To mitigate the influence of the attackedcc sensor on the system's performance. Set $\Delta\hat{x}_j(k+1)$ and w_{ij} to zero, and further calculate the optimal fusion matrix for information fusion.

Lemma 4.2: In Case 2, some of the attacked sensor coefficient matrices are set to zero, and the optimal fusion matrix can be obtained by using the Lagrange multiplier method. The matrix is as follows:

$$W_{ij}(k+1) = P_{ij}^{-1}(k+1)e(e^T P_{ij}^{-1}(k+1)e)^{-1} \quad (17)$$

where $e = (I_n, I_n, \dots, I_n)_{n|N_i| \times n}^T$.

Proof: Transform $w_{i1} + w_{i2} + \cdots + w_{iN_i} = I_n$ into

$$W_{ij}^T(k+1)e = I_n \quad (18)$$

to facilitate computation.

The performance index $J = \text{tr}(\Xi)$ is defined and (16) is substituted into J , i.e.

$$J = \text{tr}(W_{ij}(k+1)P_{ij}(k+1)W_{ij}^T(k+1)) \quad (19)$$

The problem is to solve the optimal fusion coefficient matrix by minimising the performance index under the constraint condition of (18). The introduced auxiliary functions are as follows:

$$F = J + 2\text{tr}(\Lambda(W_{ij}^T(k+1)e - I_n)) \quad (20)$$

where $\Lambda = (\lambda_{ij})$ is an n -dimensional matrix and note that $P_{ij}^T(k+1) = P_{ij}(k+1)$ in (12). The optimal fusion coefficient matrix $W_{ij}(k+1)$ is assumed.

Then, the following conditions can be obtained by applying the Karush-Kuhn-Tucker condition:

$$\frac{\partial F}{\partial W_{ij}(k+1)} = P_{ij}(k+1)W_{ij}(k+1) + e\Lambda = 0 \quad (21)$$

The matrix equation obtained by combining (18) and (21) is

$$\begin{bmatrix} P_{ij}(k+1) & e \\ e^T & 0 \end{bmatrix} \begin{bmatrix} W_{ij}(k+1) \\ \Lambda \end{bmatrix} = \begin{bmatrix} 0 \\ I_n \end{bmatrix} \quad (22)$$

The application of the inverse matrix calculation formula leads to the conclusion that

$$\begin{aligned} \begin{bmatrix} W_{ij}(k+1) \\ \Lambda \end{bmatrix} &= \begin{bmatrix} P_{ij}(k+1) & e \\ e^T & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ I_n \end{bmatrix} \\ W_{ij}(K+1) &= P_{ij}(K+1)^{-1}e(e^T P_{ij}^{-1}(K+1)e)^{-1} \end{aligned} \quad (23)$$

This proof is completed. ■

4.4. Deduction of fusion effect

The fusion algorithm undergoes rigorous theoretical performance assessment. The parameter $\alpha_{ij}(k)$ is defined as the estimation error caused solely by network effects and noise when the j th sensor remains unattacked. The estimation error at the fusion center can be computed under normal operation of all sensors.

Lemma 4.3 (Suo et al., 2024a): If the attack signal $z_{ij}(k) = 0$, the fusion state estimation error at the fusion

center satisfies

$$|\hat{x}_i^{rw}(k) - \hat{x}_i(k)| \leq \|\alpha_{ij}(k)\|_\infty \quad (24)$$

where $\hat{x}_i^{rw}(k)$ is the fusion state estimation without attack, $\|\alpha_{ij}(k)\|_\infty = \max_{j \in N_i} \{\|\alpha_{ij}(k)\|_\infty\}$, and satisfies that $\|\alpha_{ij}(k)\|_\infty \leq \frac{1}{|N_i|} \sum_{j \in N_i} \|\alpha_{ij}(k)\|_\infty$.

The noteworthy aspect is that when the j th sensor is attacked, $\alpha_{ij}(k)$ is the estimation error primarily caused by the attack signal rather than network effects and sensor noise. Under partial compromise of the fusion center's adjacent sensors, the proposed algorithm rigorously confines the state estimation error within bounded limits.

Theorem 4.2: Based on Lemma 4.3, the fusion center's state estimation error remains bounded when adversarial sensors do not exceed the prescribed threshold, i.e.

$$|\hat{x}_i^{ra}(k) - \hat{x}_i(k)| \leq 3 \|\alpha_{ij}(k)\|_\infty \quad (25)$$

where $\hat{x}_i^{ra}(k)$ is the fusion state estimation under attack.

Proof: When $z_{ij}(k) \neq 0$, the sensors in set $M_i \subseteq N_i$ have been subjected to attacks.

$$\begin{aligned} |\hat{x}_i^{ra}(k) - \hat{x}_i(k)| &= |\hat{x}_i^{ra}(k) - \hat{x}_i^{rw}(k) + \hat{x}_i^{rw}(k) - \hat{x}_i(k)| \\ &\stackrel{(a)}{\leq} |\hat{x}_i^{ra}(k) - \hat{x}_i^{rw}(k)| + |\hat{x}_i^{rw}(k) - \hat{x}_i(k)| \\ &\stackrel{(b)}{\leq} |\hat{x}_i^{ra}(k) - \hat{x}_i^{rw}(k)| + \|\alpha_{ij}(k)\|_\infty \\ &= \left| \sum_{j \in (N_i/M_i)} W_{ij}(k) \hat{x}_{ij}(k) + \sum_{j \in M_i} W_{ij}(k) \hat{x}_{ij}(k) \right. \\ &\quad \left. - \sum_{j \in N_i} W_{ij}(k) \hat{x}_{ij}(k) \right| + \|\alpha_{ij}(k)\|_\infty \\ &= \left| \sum_{j \in (N_i/M_i)} (W_{ij}(k) - W_{ij}(k)) \hat{x}_{ij}(k) \right. \\ &\quad \left. - \sum_{j \in M_i} W_{ij}(k) \hat{x}_{ij}(k) \right| + \|\alpha_{ij}(k)\|_\infty \\ &\stackrel{(c)}{\leq} \left| \sum_{j \in (N_i/M_i)} \left(\frac{1}{|N_i/M_i|} I_n - W_{ij}(k) \right) \hat{x}_{ij}(k) \right. \\ &\quad \left. - \sum_{j \in M_i} W_{ij}(k) \hat{x}_{ij}(k) \right| + \|\alpha_{ij}(k)\|_\infty \end{aligned}$$

$$\begin{aligned} &= \left| \sum_{j \in (N_i/M_i)} \frac{1}{|N_i/M_i|} \hat{x}_{ij}(k) - \hat{x}_i(k) + \hat{x}_i(k) \right. \\ &\quad \left. - \sum_{j \in M_i} W_{ij}(k) \hat{x}_{ij}(k) \right| + \|\alpha_{ij}(k)\|_\infty \\ &\stackrel{(d)}{\leq} \left| \sum_{j \in (N_i/M_i)} \frac{1}{|N_i/M_i|} \hat{x}_{ij}(k) - \hat{x}_i(k) \right| \\ &\quad + \left| \hat{x}_i(k) - \sum_{j \in M_i} W_{ij}(k) \hat{x}_{ij}(k) \right| + \|\alpha_{ij}(k)\|_\infty \\ &\stackrel{(e)}{\leq} \|\alpha_{ij}(k)\|_\infty + \|\alpha_{ij}(k)\|_\infty + \|\alpha_{ij}(k)\|_\infty \end{aligned}$$

This proof is completed. ■

Inequalities (a) and (d) follow directly from the triangular inequality, while (b) is derived via Lemma 4.2. Inequality (c) stems from the fundamental property that the harmonic mean is bounded by the arithmetic mean, coupled with the zero-matrix configuration of the compromised sensor's coefficient matrix. The validity of Inequality (e) is supported by Lemma 4.2.

Remark 4.5: When the number of attacked sensors adjacent to the fusion center satisfies $|M_i| \leq s$, the fusion algorithm ensures that the security state estimation error of the fusion center is bounded. Under the premise that the spectral radius of the system is less than 1, the fusion algorithm maximises the fault tolerance of the attack through its unique detection and reconstruction mechanism. Therefore, the algorithm shows great application potential for precision tracking systems that have met or can meet the stability conditions with minor corrections.

5. Simulation example

This section considers using a radar tracking system (Sun & Deng, 2004) with multiple sensors to simulate the above results. The system model is as follows:

$$x(k+1) = \begin{bmatrix} 1 & Tk & T^2k/2 \\ 0 & 1 & Tk \\ 0 & 0 & 1 \end{bmatrix} x(k) + \omega(k) \quad (26)$$

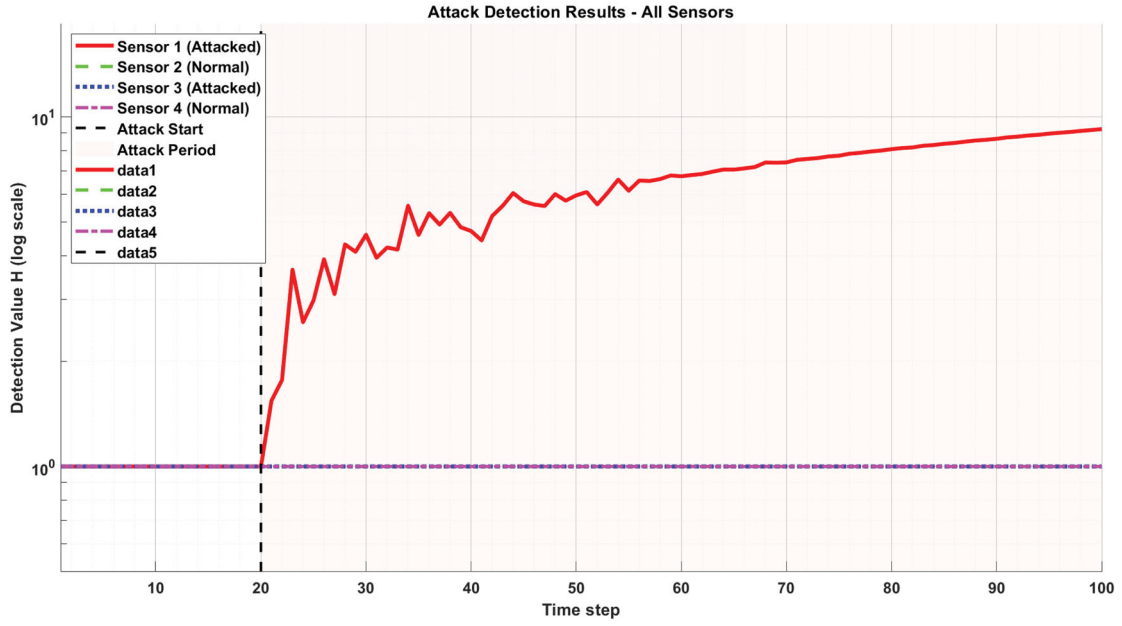


Figure 2. $\Gamma_{ij}^a(k) / \Gamma_{ij}^w(k)$ of the first sensor. The attacker launches attacks on the first sensor at $k = 20$.

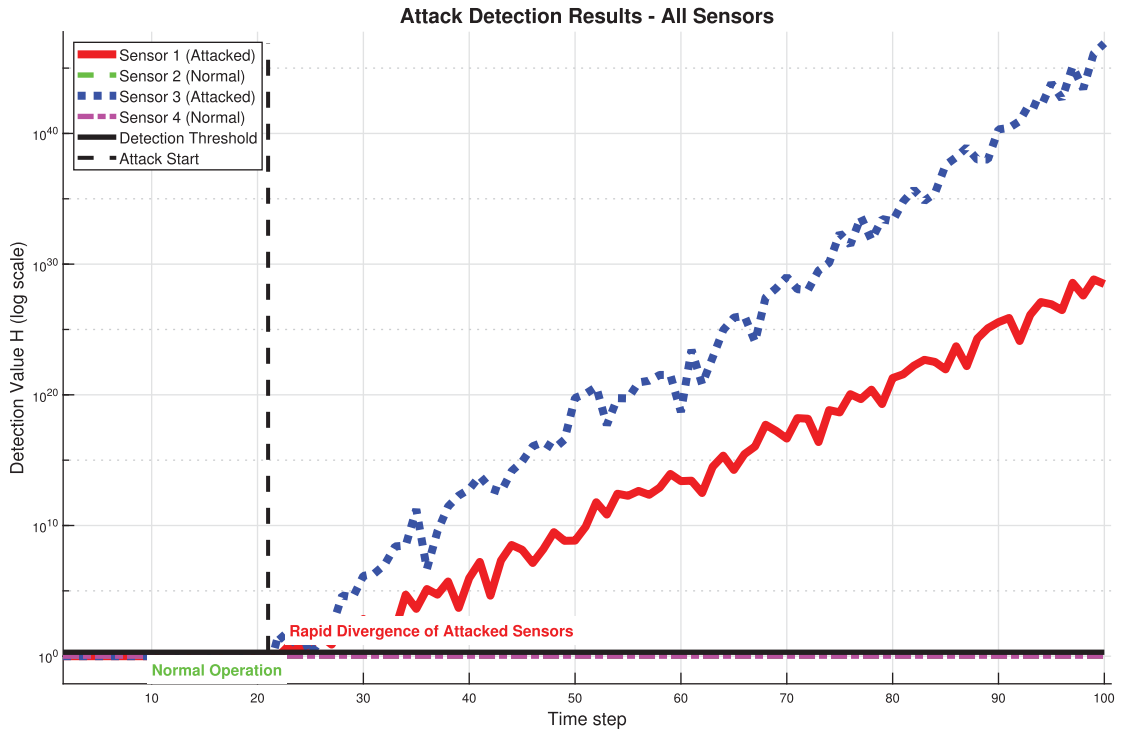


Figure 3. $\Gamma_{ij}^a(k) / \Gamma_{ij}^w(k)$ for the first and third sensors. The attacker attacks the first and third sensors at $k = 20$.

where T represents the sampling time., the state is $x(k) = [s(k) \ v(k) \ a(k)]^T$, where $s(k)$, $v(k)$, and $a(k)$ represent the position, velocity, and acceleration of the system at time k , respectively. And the initial state is $x(0) = [0 \ 0 \ 0]^T$.

The assumption is made that the sensor located around the fusion center is capable of measuring the

signal in the system state, and the measurement model is $y_i(k) = C_i x(k) + v_i(k)$. Their measurement noise $v_i(k)$ obeys $v_i(k) \sim \mathcal{N}(0, \sigma_j^2)$, where $\sigma_j^2 = 0.5$ and $T = 0.01$. This configuration deploys five sensors for observation, with sensor $i = 5$ positioned as the fusion center and its neighbouring sensors comprising the set $N_5 = 1, 2, 3, 4$. The observation matrix of the fusion

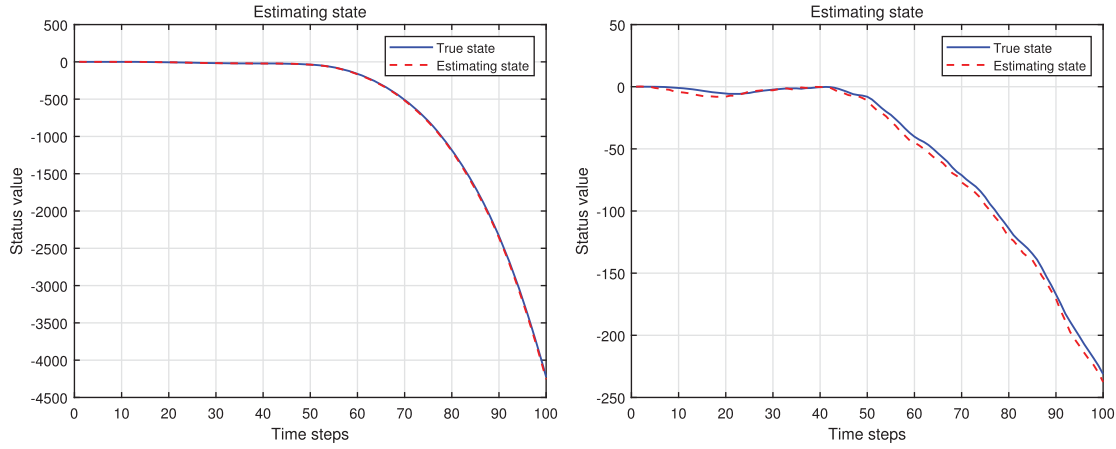


Figure 4. The state estimation of the system position $s(k)$ and velocity $v(k)$ obtained by the fusion algorithm.

center is $C_5=[0,0,1]$, and the observation matrix of other sensors are shown in the following. $C_1=[1,0,0]$, $C_2=[0,1,0]$, $C_3=[0,0,1]$, $C_4=[0.6,0,0]$.

Firstly, during the normal operation of the system, the Γ_{ij}^w when the system is not attacked is calculated. Then, the attack signal is injected at time $k = 20$. The change results of the detector corresponding to each sensor can be obtained in Figures 2 and 3. In this detection algorithm, $\Gamma_{ij}^a(k)/\Gamma_{ij}^w(k)$ is used as the detection metric. When the sensor is running normally, The theoretical value $\Gamma_{ij}^a(k)/\Gamma_{ij}^w(k)$ of the index is always 1. However, when a sensor attack occurs, the index will increase according to Theorem 3.1. Considering that Sensor $i = 1$ is attacked and Sensor $i = 1$ and Sensor $i = 3$ are attacked at the same time, the FDI attack signal is mainly used to evaluate the effectiveness of the detection mechanism. In order to more clearly reflect the effect of the detector as shown in Figures 2 and 3, the detection results of different sensors are represented by different types of curves.

Finally, based on the results of the above detection algorithm, this paper employs the s -sparse method to deal with the attacked sensor network. The fusion algorithm originally used in Sun and Deng (2004) is reconfigured, and the fusion estimation accuracy is improved by sparse processing of the detected attacked sensors. Since the network attack signal is initiated by the attacker and is different from the fault, it is very important to design an estimation model to defend against such attack signals in these systems. The estimation performance of the fusion algorithm is demonstrated in Figures 4 and 5. The estimation performance of acceleration $a(k)$ is clearly superior in comparison with (Sun & Deng, 2004). Moreover, the robustness of

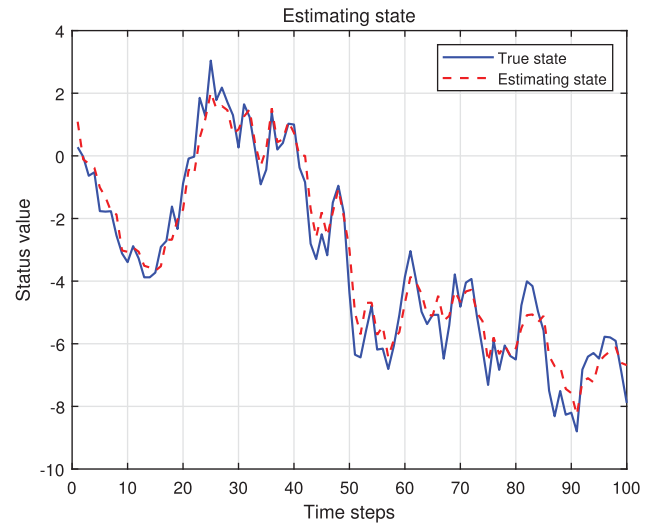


Figure 5. The change of acceleration $a(k)$ in the model and the effect of fusion state estimation.

the fusion algorithm is further emphasised through the injection of network attacks.

6. Conclusions

This manuscript focuses on adversarial threat detection and secure state estimation in sensor networks from the perspective of defensive strategies. The focus is on the vulnerability of communication channels between sensors in a multi-sensor network under attack. An innovative attack detection algorithm based on AOEP is proposed, and simulation results show its effectiveness in promptly detecting attack signals and issuing timely alarms. The attacked sensor undergoes sparse processing, followed by state estimation

through fusion. The optimal fusion estimate is determined using the Lagrange multipliers method, minimising estimation error at the fusion center and calculating the optimal fusion coefficient matrix. Finally, the unified framework design of fusion algorithm is realised.

Furthermore, while the proposed framework is designed for generic FDI attacks, its efficacy against advanced threat models warrants further investigation. For example, for well-designed stealth attacks and mixed attacks that evade statistical detection, the performance of the current detector depends on the energy profile of the attack and may have limitations. Because the s-sparse observability of the system provides basic tolerance for a finite number of faulty or unavailable sensors. Exploring this robust form boundary and developing countermeasures against perfect stealth attacks are important directions for future work.

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Data availability statement

Data sharing is not applicable to this article as no new data were created or analysed in this study.

References

- Ali, S., Al Balushi, T., Nadir, Z., & Hussain, O. K. (2018). *Cyber security for cyber physical systems* (Vol. 768). Springer.
- Chen, M., Ge, S., & Ren, B. (2011). Adaptive tracking control of uncertain MIMO nonlinear systems with input constraints. *Automatica*, 47(3), 452–465. <https://doi.org/10.1016/j.automatica.2011.01.025>
- Chen, Y., & Li, Y. (2021). Diffusion event-triggered sequential asynchronous state estimation algorithm for stochastic multiplicative noise systems. *International Journal of Systems Science*, 53(1), 122–137. <https://doi.org/10.1080/00207721.2021.1939192>
- Chen, G., Zhang, Y., Gu, S., & Hu, W. (2022). Resilient state estimation and control of cyber-physical systems against false data injection attacks on both actuator and sensors. *IEEE Transactions on Control of Network Systems*, 9(1), 500–510. <https://doi.org/10.1109/TCNS.2021.3113265>
- Du, D., Li, X., Li, W., Chen, R., Fei, M., & Wu, L. (2019). Admm-based distributed state estimation of smart grid under data deception and denial of service attacks. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 49(8), 1698–1711. <https://doi.org/10.1109/TSMC.6221021>
- Fang, C. (2021). *Research on detection and defense of data integrity attacks in cyber-physical systems*. Zhejiang Univ.
- Lin, H.-T. (2013). Implementing smart homes with open source solutions. *International Journal of Smart Home*, 7(4), 289–295.
- Liu, X., Deng, F., Zeng, P., Gao, X., & Zhao, X. (2022). Sampled-data resilient control for stochastic nonlinear CPSs under DoS attacks. *International Journal of Systems Science*, 54(5), 1165–1171. <https://doi.org/10.1080/00207721.2022.2158445>
- Liu, W., Sun, J., Wang, G., Bullo, F., & Chen, J. (2023). Data-driven resilient predictive control under denial-of-service. *IEEE Transactions on Automatic Control*, 68(8), 4722–4737. <https://doi.org/10.1109/TAC.2022.3209399>
- Lu, A., & Yang, G. (2020). Observer-based control for cyber-physical systems under Denial-of-service with a decentralized event-triggered scheme. *IEEE Transactions on Cybernetics*, 50(12), 4886–4895. <https://doi.org/10.1109/TCYB.6221036>
- Lu, A., & Yang, G. (2023). Detection and identification of sparse sensor attacks in cyber physical systems with side information. *IEEE Transactions on Automatic Control*, 68(9), 5349–5364. <https://doi.org/10.1109/TAC.2022.3218545>
- Luo, X., Fang, C., He, J., Zhao, C., & Paccagnan, D. (2024). Model-free false data injection attack in networked control systems: A feedback optimization approach. *IEEE Transactions on Automatic Control*, 69(4), 1234–1246.
- Mishra, S., Shoukry, Y., Karamchandani, N., Diggavi, S. N., & Tabuada, P. (2017). Secure state estimation against sensor attacks in the presence of noise. *IEEE Transactions on Control of Network Systems*, 4(1), 49–59. <https://doi.org/10.1109/TCNS.2016.2606880>
- Shen, J., Weng, P., Shen, Y., Chen, B., & Yu, L. (2023). Distributed fusion estimation for nonlinear cyber-physical systems with attacked control signals. *IEEE Systems Journal*, 17(1), 1216–1223. <https://doi.org/10.1109/JSYST.2022.3151713>
- Shi, D., Elliott, R. J., & Chen, T. (2017). On finite-state stochastic modeling and secure estimation of cyber-physical systems. *IEEE Transactions on Automatic Control*, 62(1), 65–80. <https://doi.org/10.1109/TAC.2016.2541919>
- Shoukry, Y., Nuzzo, P., Puggelli, A., Sangiovanni-Vincentelli, A. L., Seshia, S. A., & Tabuada, P. (2017). Secure state estimation for cyberphysical systems under sensor attacks: A satisfiability modulo theory approach. *IEEE Transactions on Automatic Control*, 62(10), 4917–4932. <https://doi.org/10.1109/TAC.2017.2676679>
- Su, B., Dai, S., Xi, Q., & Han, X. (2023). Research on state estimation algorithm of Maze robot based on multi-sensor fusion in complex environment. In *China Automation Congress (CAC)* (pp. 189–195). Chongqing, China.
- Sun, S., & Deng, Z. (2004). Multi-sensor optimal information fusion Kalman filter. *Automatica*, 40(6), 1017–1023. <https://doi.org/10.1016/j.automatica.2004.01.014>

- Sun, L., Wu, T., & Zhang, Y. (2023). A defense strategy for false data injection attacks in multi-agent systems. *International Journal of Systems Science*, 54(16), 3071–3084. <https://doi.org/10.1080/00207721.2023.2268239>
- Suo, Y., Chai, R., Chai, S., Farhan, I. M., Xia, Y., & Liu, G.-P. (2024a). Attack detection and secure state estimation of collectively observable cyber-physical systems under false data injection attacks. *IEEE Transactions on Automatic Control*, 69(3), 2067–2074. <https://doi.org/10.1109/TAC.2023.3316160>
- Suo, Y., Chai, S., Chai, R., Pang, Z.-H., Xia, Y., & Liu, G.-P. (2024b). Security defense of large-scale networks under false data injection attacks: An attack detection scheduling approach. *IEEE Transactions on Information Forensics and Security*, 18, 1228–1241.
- Wang, Y., Liu, B., & Chen, J. (2023). Attacks detection against two-channel partial FDI in CPS: A moving target defense method. In *China Automation Congress(CAC)* (pp. 314–319). Chongqing, China.
- Wang, L., Yan, H. Wang, & Shi, Y. (2024). Event-triggered bipartite consensus tracking for multiagent systems under DoS attacks. *International Journal of Systems Science*, 56(5), 1113–1129. <https://doi.org/10.1080/00207721.2024.2412188>
- Xiong, G., Zhu, F., Liu, X., & Dong, X. (2015). Cyber-physical-social system in intelligent transportation. *IEEE/CAA Journal of Automatica Sinica*, 2(3), 320–333. <https://doi.org/10.1109/JAS.2015.7152667>
- Ye, D., & Zhang, T.-Y. (2020). Summation detector for false data-injection attack in cyber-physical systems. *IEEE Transactions on Cybernetics*, 50(6), 2338–2345. <https://doi.org/10.1109/TCYB.6221036>
- Yi, N., & Xu, J. (2024). Defense strategy selection based on incomplete information game for the false data injection attack. *International Journal of Systems Science*, 55(14), 2897–2913. <https://doi.org/10.1080/00207721.2024.2363546>
- Zhao, Z., Xu, Y., Li, Y., Zhen, Z., Yang, Y., & Shi, Y. (2023). Data-driven attack detection and identification for cyber-physical systems under sparse sensor attacks. *IEEE Transactions on Automatic Control*, 68(10), 6330–6337. <https://doi.org/10.1109/TAC.2022.3230360>
- Zhao, Y., & Zhu, F. (2021). Security control of cyber-physical systems under Denial-of-Service sensor attack: A switching approach. In *IEEE 10th Data Driven Control and Learning Systems Conference(DDCLS)* (pp. 1112–1117). Suzhou, China.